

*Non-Calculator***Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.****You have 30 minutes.**_____
Signature**SHOW ALL WORK FOR FULL CREDIT!**

Find the product. Write the product in rectangular form, using exact values. (3 points)

1. $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

Find the quotient. Write it in rectangular form, using exact values. (3 points)

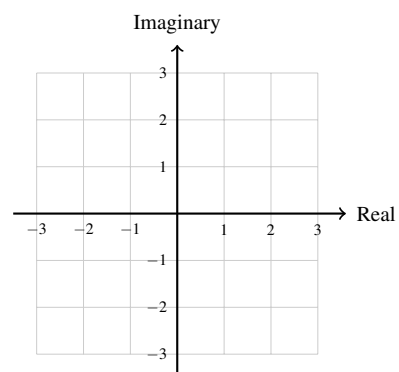
2. $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

Find the power. Write the answer in rectangular form, using exact values. (3 points)

3. $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

4. $-1 - i\sqrt{3}$



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

5. a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

ANSWER KEY – Non-Calculator

SHOW ALL WORK FOR FULL CREDIT!*Every line shown. Point splits are an estimate of partial credit (halves are possible).*

Find the product. Write the product in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \right) \right] \left[\sqrt{2} \left(\cos \frac{11\pi}{12} + i \sin \frac{11\pi}{12} \right) \right]$

1. Multiply the r 's, add the θ 's (common denominator first).

$$r = \sqrt{2} \cdot \sqrt{2} = 2 \quad \theta = \frac{5\pi}{4} + \frac{11\pi}{12} = \frac{15\pi}{12} + \frac{11\pi}{12} = \frac{26\pi}{12} = \frac{13\pi}{6} \xrightarrow{-2\pi} \frac{\pi}{6}$$

$$2 \left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} + i \left(\frac{1}{2} \right) \right) = \boxed{\sqrt{3} + i}$$

Trap: given in radians, stay in radians. $2\pi = \frac{12\pi}{6}$.

Find the quotient. Write it in rectangular form, using exact values. (3 points)

Problem: $\frac{\sqrt{2} + i\sqrt{6}}{-\sqrt{3} - i}$

2. Convert top and bottom, divide the r 's, subtract the θ 's.

$$\text{Top: } r = \sqrt{(\sqrt{2})^2 + (\sqrt{6})^2} = \sqrt{8} = 2\sqrt{2} \quad \tan \theta = \frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3} \Rightarrow \text{QI, } 60^\circ$$

$$\text{Bottom: } r = \sqrt{(-\sqrt{3})^2 + (-1)^2} = 2 \quad \tan \theta = \frac{-1}{-\sqrt{3}} = \frac{1}{\sqrt{3}} \Rightarrow \text{ref } 30^\circ, \text{ QIII} \Rightarrow 210^\circ$$

$$\frac{2\sqrt{2} \text{ cis } 60^\circ}{2 \text{ cis } 210^\circ} = \sqrt{2} \text{ cis } (-150^\circ) \xrightarrow{+360^\circ} \sqrt{2} \text{ cis } 210^\circ = \sqrt{2} \left(-\frac{\sqrt{3}}{2} + i \left(-\frac{1}{2} \right) \right) = \boxed{-\frac{\sqrt{6}}{2} - \frac{\sqrt{2}}{2} i}$$

Trap: both parts of the bottom negative means QIII: 210° , not 30° .

Find the power. Write the answer in rectangular form, using exact values. (3 points)

Problem: $\left[\sqrt{2} \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \right]^4$

3. De Moivre: raise r to the 4th, multiply the angle by 4.

$$(\sqrt{2})^4 = 4 \quad 4 \cdot \frac{2\pi}{3} = \frac{8\pi}{3} \xrightarrow{-2\pi} \frac{8\pi}{3} - \frac{6\pi}{3} = \frac{2\pi}{3}$$

$$4 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = 4 \left(-\frac{1}{2} + i \left(\frac{\sqrt{3}}{2} \right) \right) = \boxed{-2 + 2\sqrt{3}i}$$

Trap: $(\sqrt{2})^4 = 4$, not $4\sqrt{2}$. $\frac{2\pi}{3}$ is QII: cos negative, sin positive.

Find the fifth roots of the number. Leave answers in trig form with θ in radians. Graph each root as a vector in the complex plane. (3 points)

Problem: $-1 - i\sqrt{3}$

4. Convert, then roots. Radians throughout.

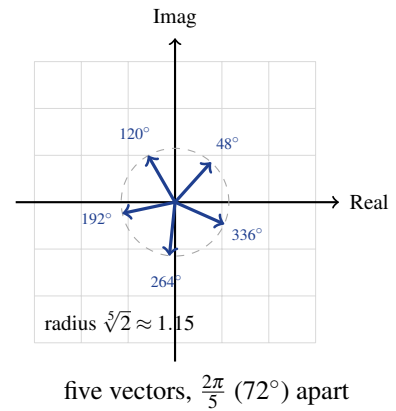
$$r = \sqrt{(-1)^2 + (-\sqrt{3})^2} = \sqrt{4} = 2$$

$$\tan \theta = \frac{-\sqrt{3}}{-1} = \sqrt{3} \Rightarrow \text{ref } \frac{\pi}{3}, \text{ QIII} \Rightarrow \frac{4\pi}{3}$$

$$\sqrt[5]{2} \quad \text{first: } \frac{4\pi/3}{5} = \frac{4\pi}{15} \quad + \frac{2\pi}{5} = + \frac{6\pi}{15}$$

$$\sqrt[5]{2} \text{ cis } \frac{4\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{2\pi}{3}, \sqrt[5]{2} \text{ cis } \frac{16\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{22\pi}{15}, \sqrt[5]{2} \text{ cis } \frac{28\pi}{15}$$

Trap: $\frac{10\pi}{15} = \frac{2\pi}{3}$ (simplify). $\sqrt[5]{2}$ stays a radical.



One of the fourth roots of a complex number w is $-\sqrt{2} + i\sqrt{2}$. (3 points)

Problem: a) Determine the rectangular form of its other three fourth roots.

b) Find w in rectangular form.

5. All n th roots have the same r and sit $\frac{360^\circ}{n}$ apart.

$$-\sqrt{2} + i\sqrt{2}: r = \sqrt{2+2} = 2 \quad \tan \theta = -1, \text{ QII} \Rightarrow 135^\circ \quad + 90^\circ$$

$$\text{a) } 2 \text{ cis } 225^\circ = -\sqrt{2} - i\sqrt{2} \quad 2 \text{ cis } 315^\circ = \sqrt{2} - i\sqrt{2} \quad 2 \text{ cis } 45^\circ = \sqrt{2} + i\sqrt{2}$$

$$\text{b) } w = (2 \text{ cis } 135^\circ)^4 = 16 \text{ cis } 540^\circ \xrightarrow{-360^\circ} 16 \text{ cis } 180^\circ = 16(-1) = -16$$

$$\text{a) } -\sqrt{2} - i\sqrt{2}, \sqrt{2} - i\sqrt{2}, \sqrt{2} + i\sqrt{2} \quad \text{b) } w = -16$$

Trap: fourth roots are 90° apart (cube roots are 120°). To get w , raise the root to the 4th.